Design and Model Summary Report

Design Overview

DESIGN	RUNS	REPLICATES
Central Composite	**9**	**0**

CENTER POINTS	CONTINUOUS
1	**2**

Process Factors

Factor	Type	Settings	Coded levels
Load	Continuous	5 to 15	-1, 0, 1
Classifier	Continuous	1 to 3	-1, 0, 1

Design matrix
The complete design matrix in actual factor units is provided in Appendix A.

Model Structure

Responses: Y1, Y2

Included Terms

Intercept	Yes
Linear Terms	Load, Classifier
2-Term Int.	Load : Classifier
Quadratic Terms	Load^2, Classifier^2
3-Term Int.	Not included

Model Matrix Diagnostics

Metric	Value
Observations	9
Columns	6
Rank	6
Aliased columns	0
Intercept	Yes
Regression DF	5
Residual DF	3
Status	Full rank

Full-rank model
All model-matrix columns are linearly independent.

General Diagnostics

Variance Inflation Factors (VIF)

VIF < 5 is low, 5-10 is elevated, and VIF > 10 indicates high multicollinearity.

Term	VIF	Flag
Load	1	Low
Classifier	1	Low
Load:Classifier	1	Low
Load^2	1	Low
Classifier^2	1	Low

Leverage

Average leverage: 0.6667

Run	Idx	Leverage
1	1	0.8056
2	2	0.8056
3	3	0.8056
4	4	0.8056
5	5	0.5556
6	6	0.5556
7	7	0.5556
8	8	0.5556
9	9	0.5556

Response Conditions

Optimization limits and goals configured for each response.

Response	Lower Limit	Upper Limit	Goal
Y1	70	No	Maximize
Y2	No	1	Minimize

Response Analysis: Y1

Model Metrics

R^2	Adj. R^2	Q^2 (LOO)	PRESS (LOO)	RMSE	RMSE (LOO)
0.9916	0.9776	0.9013	48.721	0.6785	2.3267

Coefficient Summary

Term	Coefficient	Std. Error	95% CI half-width	Lower CI	Upper CI	p-value
Int	76.9011	0.8759	2.7875	74.1136	79.6887	3.26e-06
Load	-4.8383	0.4798	1.5268	-6.3651	-3.3115	0.0021
Classifier	-5.6	0.4798	1.5268	-7.1268	-4.0732	0.0014
Load:Classifier	1.835	0.5876	1.8699	-0.0349	3.7049	0.0524
Load^2	-1.7117	0.831	2.6445	-4.3562	0.9328	0.1315
Classifier^2	-8.4167	0.831	2.6445	-11.0612	-5.7722	0.0021

Model F-test

F-value	F-crit (95%)	F-crit (99%)	p-value
70.9087	9.0135	28.2371	0.0026

Highly significant
At the 1% level, there is evidence that the model explains variation beyond a constant response (p = 0.00259).

Lack-of-Fit F-test

Not applicable
No replicated-run lack-of-fit test is applicable.

ANOVA

SD = sqrt(MS). Only the Residuals row reports residual standard deviation; RMSE = sqrt(SS_res / n).

Source	SS	dof	MS	SD
Total	493.7689	8	61.7211	7.8563
Regression	489.6259	5	97.9252	9.8957
Residuals	4.143	3	1.381	1.1752

Replicate Summary

Not applicable
No replicated runs are present.

Observed and Predicted Responses

Run	Idx	Observed	Predicted
1	1	79	79.0461
2	2	63.5	64.1761
3	3	66.58	65.6994
4	4	58.42	58.1694
5	5	69.22	70.3511

Run	Idx	Observed	Predicted
6	6	80.75	80.0278
7	7	63.31	62.8844
8	8	73.25	74.0844
9	9	77.31	76.9011

Response Analysis: Y2

Model Metrics

R^2	Adj. R^2	Q^2 (LOO)	PRESS (LOO)	RMSE	RMSE (LOO)
0.9941	0.9842	0.9318	0.1306	0.0355	0.1204

Coefficient Summary

Term	Coefficient	Std. Error	95% CI half-width	Lower CI	Upper CI	p-value
Int	0.1467	0.0458	0.1458	0.000872	0.2925	0.0493
Load	0.455	0.0251	0.0799	0.3751	0.5349	0.000366
Classifier	-0.2133	0.0251	0.0799	-0.2932	-0.1335	0.0034
Load:Classifier	-0.23	0.0307	0.0978	-0.3278	-0.1322	0.0049
Load^2	0.285	0.0435	0.1383	0.1467	0.4233	0.0072
Classifier^2	0.08	0.0435	0.1383	-0.0583	0.2183	0.1629

Model F-test

F-value	F-crit (95%)	F-crit (99%)	p-value
100.6976	9.0135	28.2371	0.0015

Highly significant
At the 1% level, there is evidence that the model explains variation beyond a constant response (p = 0.00154).

Lack-of-Fit F-test

Not applicable
No replicated-run lack-of-fit test is applicable.

ANOVA

SD = sqrt(MS). Only the Residuals row reports residual standard deviation; RMSE = sqrt(SS_res / n).

Source	SS	dof	MS	SD
Total	1.9134	8	0.2392	0.4891
Regression	1.9021	5	0.3804	0.6168
Residuals	0.0113	3	0.0038	0.0615

Replicate Summary

Not applicable
No replicated runs are present.

Observed and Predicted Responses

Run	Idx	Observed	Predicted
1	1	0.05	0.04
2	2	0.03	0.0733
3	3	1.44	1.41
4	4	0.5	0.5233
5	5	0.88	0.8867

Run	Idx	Observed	Predicted
6	6	0.01	-0.0233
7	7	0.08	0.0133
8	8	0.4	0.44
9	9	0.12	0.1467

Appendix A - Design Matrix

Run	Idx	Load	Classifier
1	1	5	1
2	2	5	3
3	3	15	1
4	4	15	3
5	5	15	2
6	6	5	2
7	7	10	3
8	8	10	1
9	9	10	2